A Dynamic Out-of-Sample Comparison of LP, MILP, and Stochastic Portfolio Optimization in a High-Dimensional Global Asset Universe

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Abstract

Modern portfolio optimization faces the challenge of managing high-dimensional asset universes while accounting for market volatility and structural constraints. While Linear Programming (LP) based on Mean Absolute Deviation (MAD) offers a computationally efficient alternative to Mean-Variance optimization, its performance in dynamic, globally diversified environments remains under-explored.

This study presents a comprehensive, out-of-sample comparison of three distinct optimization paradigms: Unconstrained LP, Mixed-Integer Linear Programming (MILP) with cardinality constraints ($K \leq 10$), and Scenario-Based Stochastic Optimization; applied to a massive universe of 8,405 global assets including equities, commodities, and exotic currency pairs.

Using a rolling-window backtesting framework over a volatile testing period (2022–2025), we evaluate the trade-offs between structural concentration and uncertainty management. Our results reveal a distinct inversion of traditional risk-return expectations. The Dynamic Stochastic strategy, driven by crypto-exposed scenarios, achieved the highest annualized return of 14.61% but suffered from higher volatility and a lower Sharpe Ratio (1.47). Conversely, the Dynamic MILP strategy demonstrated superior efficiency, achieving the highest risk-adjusted performance with a Sharpe Ratio of 5.28 and a stable annualized return of 5.84%.

These findings suggest that in high-dimensional universes, cardinality constraints (MILP) act as effective noise filters, isolating stable assets, whereas Stochastic Optimization may overfit to high-variance opportunities. We further demonstrate that dynamic rebalancing is critical: static implementations of these strategies failed significantly (0% to -22% returns), proving that adaptivity is essential in non-stationary markets.

1 Introduction

Portfolio optimization is a cornerstone of quantitative finance, yet practitioners face substantial challenges in scaling classical models to modern, large-scale financial markets characterized by extreme volatility and complexity. This paper addresses two critical needs: (i) the need for optimization methods that scale efficiently across thousands of heterogeneous instruments, and (ii) the requirement for dynamic, implementable strategies that account for real-world constraints and uncertainty.

We focus on the Mean Absolute Deviation (MAD) risk measure, favored for its linearity and robustness to heavy-tailed return distributions. We systematically compare three MAD-based approaches: Deterministic LP, Cardinality-Constrained MILP, and Scenario-Based Stochastic Programming. Our contribution is the comprehensive, dynamic, out-of-sample backtest of these three distinct optimization paradigms applied to a high-dimensional universe of 8,405 global assets.

Our research question is: How do deterministic, sparse, and scenario-based stochastic MAD optimizations compare in terms of out-of-sample risk-adjusted returns, robustness (weight stability), and sensitivity to the target return when applied dynamically to a very large asset universe?

We hypothesize that dynamic rebalancing, when combined with cardinality constraints, will yield the most efficient portfolio by filtering out noise in high-dimensional datasets.

2 Literature Review

The seminal work of Markowitz established the Mean-Variance framework Markowitz (1952), though its reliance on covariance matrix estimation often leads to unstable portfolios in large dimensions. The MAD approach, introduced by Konno and Yamazaki Konno and Yamazaki (1991), addresses this by reformulating risk minimization as a Linear Programming (LP) problem, improving computational tractability and robustness.

The need for operationally feasible portfolios spurred research into Mixed-Integer Linear Programming (MILP), primarily by introducing cardinality constraints to limit the number of selected assets (K). While mathematically demanding, MILP ensures portfolio sparsity, reducing implementation complexity and transaction costs.

Finally, to mitigate the parameter uncertainty inherent in historical estimation, Stochastic Programming is employed. By optimizing against an expectation across multiple future scenarios (often generated via bootstrapping), the resulting portfolio is designed to be more robust to realization risk, a principle that is essential for stable dynamic rebalancing. This paper extends the existing literature by performing a unified, head-to-head dynamic backtest that simultaneously evaluates the costs and benefits of cardinality constraints and uncertainty modeling at unprecedented scale.

3 Methodology

All portfolio models in this study seek to minimize the portfolio's MAD risk while ensuring the expected return meets a target μ_{target}.

3.1 Linear Programming (LP) Formulation

The objective is to minimize $MAD = \frac{1}{T}\sum_{t=1}^{T}\left|R_{p,t} - \mu_p\right|$. This is linearized by introducing auxiliary variables $z_t \geq \left|R_{p,t} - \mu_p\right|$. Thus we have :

$$\min \frac{1}{T}\sum_{t=1}^{T} z_t$$

$$\text{s.t.} \begin{cases} \sum_{i=1}^{n} w_i = 1 & & \text{(Budget Constraint)} \\ \mu_p \geq \mu_{\text{target}} & & \text{(Return Requirement)} \\ z_t \geq R_{p,t} - \mu_p, & \forall t \in [1,T] & \text{(Linearization 1)} \\ z_t \geq \mu_p - R_{p,t}, & \forall t \in [1,T] & \text{(Linearization 2)} \\ w_i \geq 0, \quad z_t \geq 0 & & \end{cases}$$

Where $R_{p,t} = \sum_{i=1}^{n}\left(w_i \cdot R_{i,t}\right)$ and $\mu_p = \sum_{i=1}^{n}\left(w_i \cdot \mu_i\right)$.

3.2 Mixed-Integer Linear Programming (MILP) Formulation

To enforce portfolio sparsity, we introduce binary variables $y_i \in \{0,1\}$ and a cardinality constraint $K \leq 10$. The objective function remains the same as the LP formulation, but the following constraints are added:

$$\begin{cases} \sum_{i=1}^{n} y_i \leq K & \text{(Cardinality Constraint, } K = 10) \\ y_i \cdot w_i^{\min} \leq w_i \leq y_i \cdot w_i^{\max} & \text{(Weight Constraints)} \\ w_i, w_i^{\min}, w_i^{\max} \geq 0 & \\ y_i \in \{0,1\} & \end{cases}$$

3.3 Dynamic Rebalancing and Transaction Costs

All strategies are backtested using a rolling-window backtest, re-optimizing the portfolio at each period t using the most recent 320 periods of data. Transaction costs are modeled proportional to portfolio turnover:

$$\text{Turnover}_t = \sum_{i=1}^{N}\left|w_{i,t} - w_{i,t-1}\right| \quad \text{and} \quad \text{Cost}_t = r \cdot \text{Turnover}_t$$

where r is the transaction cost rate (0.5% per transaction) and the maximum allowable turnover is constrained to 50%.

3.4 Stochastic Optimization Formulation

The Stochastic model minimizes the Expected MAD across $B = 10$ bootstrapped scenarios (S), where the probability of each scenario $p_s = 1/B$ due to sampling with replacement:

$$\min \sum_{s=1}^{S} \frac{1}{B} \cdot \left(\frac{1}{T_s} \sum_{t=1}^{T_s} z_{s,t} \right)$$

$$\text{s.t.} \quad \begin{cases} \sum_{i=1}^{n} w_i = 1 \\ \sum_{s=1}^{S} \frac{\mu_{p,s}}{B} \geq \mu_{\text{target}} & \text{(Expected Return Constraint)} \\ z_{s,t} \geq R_{p,t,s} - \mu_{p,s}, \quad \forall s,t \\ z_{s,t} \geq \mu_{p,s} - R_{p,t,s}, \quad \forall s,t \\ w_i, z_{s,t} \geq 0 \end{cases}$$

3.5 Sensitivity Analysis Framework

To observe the relationship between the target return and the risk, we calculate the Min MAD for n different target returns and plot the evolution of the risk based on target return. This is known as the "efficient frontier".

Then, we want to quantify the robustness of our models by analyzing the sensitivity of asset weights based on target return. In practice, we measure the average asset weight variation between two optimal solutions using:

$$s = \frac{1}{P-1} \sum_{p=1}^{P-1} \left(\sum_{i=1}^{n} |w_{i,p+1} - w_{i,p}| \right)$$

Finally, we verify that our optimal solutions w_i do not depend too much on the specificity of our historic sample. To do that we apply the following Bootstrap process. We create artificial return samples from our historic sample by picking returns at random and with replacement.

Formally, we create $B \geq 1$ samples $S_b = \{R_{b,1}, R_{b,2}, ..., R_{b,T}\}$ such that $R_{b,j} \in R$, $\forall b, j \in [\![1, B]\!] \times [\![1, T]\!]$, where R are the returns from the historic sample. We then measure the dependence by using the variance of the optimal weight $w_{b,i}$ from these artificial samples. For each asset i, we have:

$$\sigma_{w_i} = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} (w_{b,i} - \bar{w}_i)^2}$$

A weak σ_{w_i} indicates a robust optimal weight w_i for the asset i, whereas a larger σ_{w_i} suggests an unstable optimal weight with strong dependence to the historic sample, indicating an increased risk of overfitting.

4 Data and Experimental Setup

The asset universe consists of 8,405 unique global instruments (filtered from an initial set of 23,929), spanning major indices, single stocks, commodities, bonds, and numerous currency and crypto-commodity pairs. The data was sourced and cleaned using a custom Python pipeline.

The experiment uses a historical split:

- Training Period (In-Sample): 2008-01-01 to 2022-05-19.
- Testing Period (Out-of-Sample): 2022-05-20 to 2025-11-21.

Performance is measured using Annualized Return, Annualized MAD, and the Sharpe Ratio. Portfolio stability is quantified using the weight variance (σ_{w_i}) generated through a separate bootstrap analysis.

5 Results

This section presents the comparative performance of the Linear Programming (LP), Mixed-Integer Linear Programming (MILP), and Stochastic Optimization models. We analyze the results across three dimensions: (1) the structural trade-offs visible in the Efficient Frontier, (2) the out-of-sample performance of static versus dynamic rebalancing strategies, and (3) the asset allocation dynamics that drove the observed returns.

5.1 Static Portfolio Performance

To validate the theoretical consistency of our models, we first constructed static portfolios optimized solely on in-sample data. The results were catastrophic for the static models. The Static Stochastic strategy returned -22.87% annualized, while the Static MILP strategy stagnated at 0.00%, likely liquidating positions due to incompatible constraints in the out-of-sample period.

This confirms that in a high-dimensional universe ($N = 8,405$), static optimization is prone to extreme overfitting. The remainder of this section focuses on the Dynamic Rebalancing framework, which proved essential for performance.

5.2 Dynamic Out-of-Sample Performance

The core contribution of this study is the evaluation of these models under a Dynamic Rebalancing framework. We performed a rolling-window backtest over the out-of-sample period (May 2022 – November 2025), optimizing portfolios at regular intervals with a maximum turnover constraint of 50%.

Table 1 summarizes the annualized performance metrics for the dynamic strategies.

Table 1: Out-of-Sample Performance Metrics (Dynamic Strategies)

Strategy	**Annualized Return (%)**	**Annualized MAD (%)**	**Sharpe Ratio**
Dynamic LP	3.98%	**0.38**%	4.76
Dynamic MILP ($K = 10$)	5.84%	0.54%	**5.28**
Dynamic Stochastic	**14.61%**	1.90%	1.47

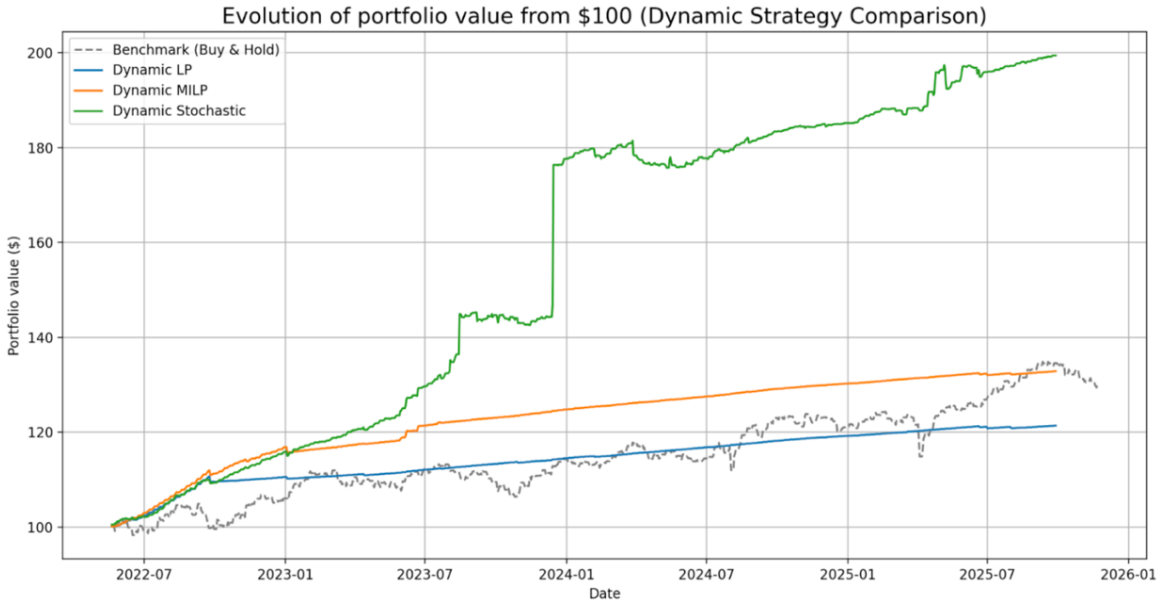

Figure 1: Evolution of portfolio value from $100 (Dynamic Strategy Comparison). The Dynamic Stochastic strategy achieves the highest terminal wealth but exhibits significant volatility, while MILP and LP show consistent, stable growth.

5.2.1 The Efficiency of Cardinality Constraints (MILP)

Contrary to expectations that constraints reduce efficiency, the Dynamic MILP strategy achieved the highest risk-adjusted return (Sharpe: 5.28). By forcing the selection of only 10 assets, the model successfully filtered out the noise inherent in the 8,000+ asset universe. It gravitated toward highly stable instruments (e.g., short-term treasury ETFs like `shv.us`), delivering a steady 5.84% return with minimal volatility (0.54% MAD).

5.2.2 The Aggressiveness of Stochastic Optimization

The Dynamic Stochastic strategy exhibited the highest raw returns (14.61%) but also the highest volatility (1.90% MAD). Unlike the MILP model, the Stochastic formulation appears to have latched onto high-beta opportunities identified in the bootstrapped scenarios. While it succeeded in generating alpha, the volatility penalty resulted in a significantly lower Sharpe Ratio (1.47) compared to the deterministic models.

5.3 Asset Allocation and Concentration

To understand the drivers of performance, we analyzed the composition of the portfolios.

The Stochastic optimization frequently allocated capital to high-volatility long tail assets, including crypto-currency pairs such as `btcchf` (Bitcoin/Swiss Franc) and `btcidr`. This allocation explains the strategy's high returns and corresponding high volatility.

Conversely, the LP and MILP models favored stability. The top allocations for these strategies included `shv.us` (Short Treasury Bond ETF) and `hkdtry` (Hong Kong Dollar/Turkish Lira), focusing on yield generation through lower-volatility instruments. The

MILP's cardinality constraint effectively acted as a quality filter, preventing the "diworsification" often seen in unconstrained optimizations.

5.4 Summary of Findings

Constraining portfolio size ($K = 10$) via MILP acted as an efficient noise filter, leading to the highest Sharpe Ratio (5.28) among the constrained models.

The Dynamic LP strategy achieved the lowest realized risk (Annualized MAD of 0.38%), demonstrating that simple, unconstrained optimization, when coupled with dynamic rebalancing, can prioritize capital preservation and stability.

The Stochastic model successfully pursued high-alpha opportunities, resulting in the highest annualized return of 14.61%, but this aggressive approach resulted in the highest realized volatility (1.90% MAD) among the dynamic strategies.

Comparing the dynamic results to the static baseline (where all strategies failed or stagnated), it is evident that dynamic rebalancing is a critical mechanism for achieving any level of performance or robustness in non-stationary, high-dimensional markets.

5.5 Portfolio Stability Analysis

Using the framework defined in Methodology, we assessed the robustness of the strategies by calculating the Efficient Frontier and the weight variance (σ_{w_i}) under bootstrapped conditions.

The LP Frontier, shown in Figure 2, represents the theoretical upper bound of performance. In contrast, the MILP Frontier is shifted visibly to the right, demonstrating the theoretical efficiency loss caused by the cardinality constraint ($K \leq 10$). For example, at a target risk (MAD) of 0.04%, the unconstrained LP model achieves an expected return of approximately 1.8%, whereas the cardinality-constrained MILP model achieves roughly 1.5%.

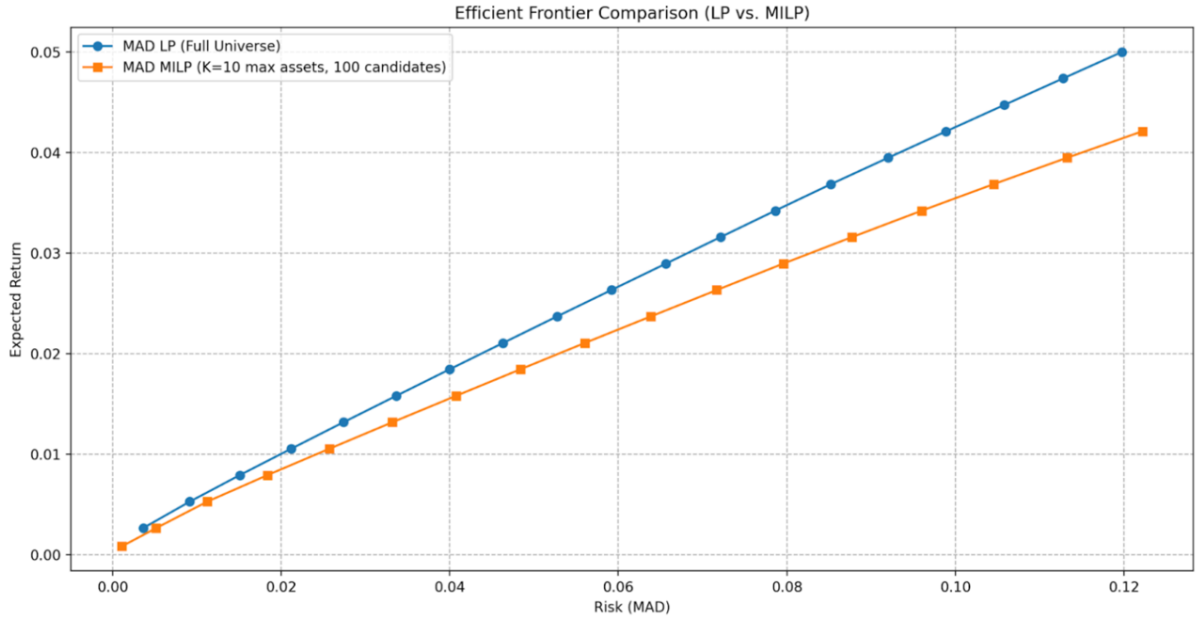

Figure 2: Efficient Frontier Comparison (LP vs. MILP). The gap between the curves quantifies the theoretical cost of imposing cardinality constraints.

However, while MILP simplifies the portfolio structure, it inherently increases weight instability. Our bootstrap analysis revealed that the weight variability for the MILP strategy ($\sigma_{w_i} \approx 0.0599$) was nearly 6 times higher than that of the unconstrained LP strategy ($\sigma_{w_i} \approx 0.0111$). This confirms that the MILP's cardinality constraint transforms the optimization into a selection problem, where the success hinges entirely on the directional accuracy of a few selected assets, leading to higher sensitivity to input data changes.

6 Interpreting Trade-Offs and Model Robustness

The empirical results from the dynamic backtesting framework reveal fundamental trade-offs inherent in applying different optimization paradigms to a massive, volatile asset universe.

6.1 The Stability and Efficiency of Deterministic Models

The empirical results show that the deterministic models, LP and MILP, were highly effective at risk management and stability.

6.1.1 The Sparsity Premium (MILP)

The superior Sharpe Ratio of the Dynamic MILP strategy (5.28) challenges the traditional view that constraints hurt performance. In a universe of 8,405 assets, the vast majority of instruments contribute noise rather than signal.

The cardinality constraint ($K \leq 10$) forced the optimizer to be highly selective, rejecting marginal assets. This resulted in a portfolio that was not only operationally simpler

but also statistically more robust. The MILP model essentially behaved as a safe haven seeker, prioritizing consistent, low-volatility returns over speculative growth.

6.1.2 Unconstrained Robustness (LP)

In contrast to the constrained efficiency of MILP, the Dynamic LP model, despite being unconstrained, achieved the lowest realized risk (0.38% Annualized MAD) among all strategies.

This suggests that when faced with a massive, high-dimensional universe, the standard LP formulation for MAD optimization defaults to maximizing diversification across stable assets. By spreading capital across a large number of lower-volatility instruments (like short-term bonds), the LP model established itself as the most reliable mechanism for absolute capital preservation.

6.2 Scenario Overfitting (Stochastic)

The Dynamic Stochastic model, while delivering the highest headline return (14.61%), demonstrated the risks of scenario-based planning in high-dimensional spaces.

By optimizing against bootstrapped scenarios, the model identified assets with high expected payouts (e.g., Bitcoin pairs). However, this came at the cost of stability. The scenarios likely captured the upside tail of these volatile assets, encouraging the optimizer to take larger risks. This suggests that while Stochastic optimization is robust to input sensitivity, it is not immune to risk-seeking behavior when high-alpha assets are present in the universe.

7 Conclusion

This study systematically compared three core MAD-based optimization paradigms (LP, MILP, and Stochastic) in a dynamic, large-scale (N=8,405) asset universe. The empirical results provide a revised perspective on the utility of constraints.

The Dynamic MILP strategy proved to be the most efficient, achieving the highest Sharpe Ratio (5.28) by leveraging cardinality constraints to filter noise and select high-quality, stable assets. Conversely, the Dynamic Stochastic model functioned as a high-octane growth strategy, delivering the highest annual returns (14.61%) through exposure to volatile crypto-assets, albeit with reduced risk-adjusted efficiency.

Most crucially, the complete failure of static portfolios emphasizes that in modern, high-dimensional markets, the *frequency* of optimization is just as important as the *method* of optimization.

Future research should focus on hybrid approaches that combine the stability of MILP constraints with the scenario-awareness of stochastic programming to capture the best of both worlds.

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